

Navigating Solvation Thermodynamics toward Thermo-electrochemical Hydrogels

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PhD Qualifying Examination Report

Sustainable Energy and Environment (SEE)

The Hong Kong University of Science and Technology (Guangzhou)

July 2026, Guangzhou

Abstract

Ionic thermoelectric materials provide a route for converting low-grade heat into electricity through thermodiffusion or thermogalvanic processes. Their performance is governed by ionic thermodynamics and transport, particularly the reaction entropy and solvation environments of redox couples. This thesis examines the molecular design of thermogalvanic electrolytes, with emphasis on solvent effects and polymer-confined hydrogel systems. For liquid ferricyanide–ferrocyanide electrolytes, the limited availability of thermopower data and the complexity of ion–solvent interactions make direct machine-learning prediction difficult. Density functional theory calculations were therefore combined with a small-data machine-learning approach for solvent screening. The resulting analysis identified solvent polarity as an important factor governing thermopower. This relationship was further examined in poly(vinyl alcohol) hydrogels. A volcano-shaped dependence of thermopower on solvent dipole moment was observed, and fine control of hydrogel polarity produced a thermopower of 3.28 mV K^{-1} . The enhancement was associated with asymmetric changes in the solvation environments of the redox ions. These findings provide a basis for relating solvent properties and redox-ion solvation to the performance of thermogalvanic electrolytes.

TABLE OF CONTENTS

Title Page	i
Abstract	ii
Table of Contents	iii
List of Figures	v
List of Tables	vi
Chapter 1	Literature Review 1
1.1	Low-Grade Heat and Thermoelectric Materials 1
1.1.1	Historical Development 2
1.1.2	Thermodiffusion and Thermogalvanic Conversion 4
1.2	Solvation Entropy Difference in Thermogalvanic Cells 5
1.2.1	Role of Solvation Entropy 7
1.2.2	Entropy Difference and Concentration Difference 7
1.2.3	From Thermopower to Recoverable Device Output 8
1.3	Redox Couples 10
1.3.1	Ferri/Ferrocyanide Redox Couple 10
1.3.2	Iron Aqua Complexes and Counterion Sensitivity 10
1.3.3	Iodide/Triiodide and Supramolecular Selectivity 11
1.3.4	Metal-Complex and Organic Redox Couples 11
1.4	Solvents and Additives 13
1.4.1	Solvent Engineering 17
1.4.2	Additives as Local Entropy Modifiers 19
1.4.3	Transport and Stability Trade-Offs 21
1.5	Hydrogel Matrices 21
1.5.1	Gelatin Systems 23
1.5.2	PVA and PAAm Systems 24
1.5.3	Cellulose and Composite Gels 25
1.5.4	Thermoresponsive and Functional Polymer Matrices 26
1.5.5	Gel Transport, Thermal Conductivity, and Device Stability 26
1.6	Research Gap 28
Chapter 2	AI-Informed Solvation Engineering of Liquid Thermogalvanic Electrolytes 30
2.1	Introduction 30
2.2	Correlation between Ion–Solvent Interactions and Thermopower 31
2.3	High-Throughput Solvent Screening 31
2.4	Experimental Validation and Model Refinement 32
2.5	Solvent Modification of a Thermosensitive Electrolyte 33

Chapter 3	Polarity-Tuned Hydrogel Electrolytes	34
	3.1 Introduction	34
	3.2 Solvent-Family Library in PVA Hydrogels	34
	3.3 Polarity-Tuned Hydrogel Electrolytes	35
	3.4 Molecular Origin of the Enhanced Thermopower	35
	3.5 Thermo-Electrochemical and Dynamic Performance	36
References		37

LIST OF FIGURES

Figure 1.1	Low-grade heat sources and the liquid-state thermocell concept, adapted from Duan et al. [1].	2
Figure 1.2	Representative evolutionary roadmap of ionic thermoelectrics, adapted from Shang et al. [2].	3
Figure 1.3	Mechanistic distinction between thermodiffusion and thermogalvanic effects in ionic thermoelectric devices, adapted from Lin et al. [3].	4
Figure 1.4	Two general routes to higher thermogalvanic thermopower: increasing the solvation-structural entropy difference and inducing a redox concentration gradient. Adapted from Duan et al. [1].	6
Figure 1.5	Material classes and representative constituents of ionic thermoelectrics, spanning liquid-state electrolytes, gel-state electrolytes, polyelectrolytes, and porous solids. Adapted from Xu et al. [4].	15

LIST OF TABLES

Table 1.1	Practical distinction between thermodiffusion-dominated and thermogalvanic ionic thermoelectrics.	5
Table 1.2	A mechanism-aware hierarchy for evaluating ionic thermoelectric materials and devices.	9
Table 1.3	Representative redox couples and reported thermopowers. Each row uses one main source for the specific value or example.	12
Table 1.4	Representative solvent and additive strategies. Solvent rows refer to changes in the bulk liquid phase; additive rows refer to selective molecular, ionic, or polymer-network interactions.	16
Table 1.5	Representative gel-matrix design principles for thermogalvanic electrolytes.	27

CHAPTER 1

LITERATURE REVIEW

This chapter examines ionic thermoelectrics through a deliberately restricted scope. It focuses on thermodiffusion (TD) and thermogalvanic (TG) conversion in liquid and quasi-solid electrolytes, with particular attention to TG redox chemistry, solvent and additive effects, and polymer-gel matrices. Ion-insertion thermoelectrochemical cycles and mixed ionic–electronic conductors are noted only where they clarify a performance metric or a device boundary. Rather than ranking materials by thermopower alone, the chapter follows a cross-scale organizing principle: molecular entropy and association set the available thermovoltage; mesoscale transport and interfaces determine how much of that voltage survives under load; and device architecture determines whether the output is continuous, cyclic, and practically recoverable [2, 4–6]. Accordingly, the discussion proceeds from low-grade heat and thermoelectric materials to TG solvation-entropy differences, redox couples, solvent and additive regulation, hydrogel matrices, and the remaining research gaps.

1.1 Low-Grade Heat and Thermoelectric Materials

Low-grade heat is widely distributed in environments where the temperature is too low for efficient recovery by conventional heat engines. It appears in industrial warm streams, data centers, microelectronics, building surfaces, and the human body. The relevant temperature difference is often only a few to several tens of kelvin, so an energy-conversion technology must operate under a small thermodynamic driving force, tolerate large areas or soft contact, and remain inexpensive enough for distributed deployment [1, 2, 7].

Electronic thermoelectric generators convert a temperature difference directly into voltage through the Seebeck effect. Their solid-state operation is compact and continuous, but high-performance electronic thermoelectric materials are often inorganic, rigid, and optimized for temperature regimes or module formats that are not ideal for near-room-temperature, flexible heat harvesting. More importantly, many electronic thermoelectrics generate Seebeck coefficients on the order of $10^2 \mu\text{V K}^{-1}$, whereas ionic thermoelectric systems can readily reach mV K^{-1} thermopowers because ions, solvent shells, and redox reactions carry much larger entropy

changes [1, 5, 6].

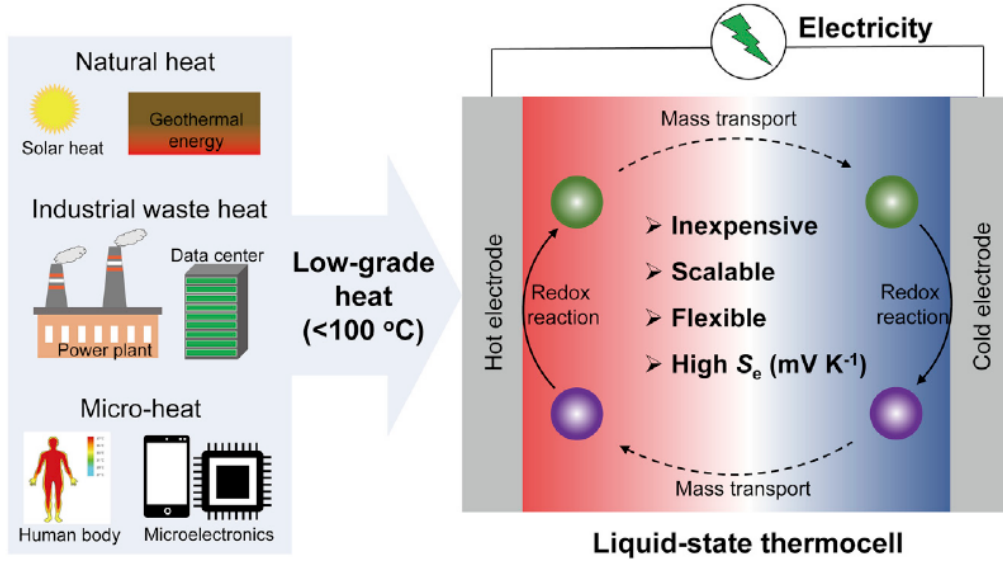


Figure 1.1: Low-grade heat sources and the liquid-state thermocell concept, adapted from Duan et al. [1].

In electrolyte-based thermoelectrics, temperature gradients are converted into electrical signals mainly through ion redistribution or redox-potential shifts. Thermodiffusion (TD) describes the Soret-driven migration of cations and anions, which can generate large thermovoltages through interfacial charge accumulation. Thermogalvanic (TG) conversion instead relies on the temperature dependence of redox equilibria at two electrodes, allowing current to be sustained when redox mass transport and electrode kinetics remain reversible. Mechanisms based on ion insertion, thermal extraction, or mixed ionic-electronic transport are not discussed further because they involve distinct electrode reactions and device architectures. The relevant contrast is therefore between TD systems, which often exhibit large but capacitor-like outputs, and TG systems, which provide a redox-based route to continuous heat-to-electricity conversion.

1.1.1 Historical Development

The development of ionic thermoelectrics can be read as a gradual broadening of what is considered an active thermoelectric material. Early thermoelectrics focused on electronic carriers in solids. The ionic thermoelectric literature then shifted attention to electrolytes, where temperature gradients affect ion distributions, solvation structures, redox potentials, and electrochemical interfaces. This shift made soft devices possible, but it also made the field more chemically complex: the measured voltage may contain contributions from thermodiffusion, redox entropy, concentration gradients, electrode kinetics, and polymer-ion interactions.

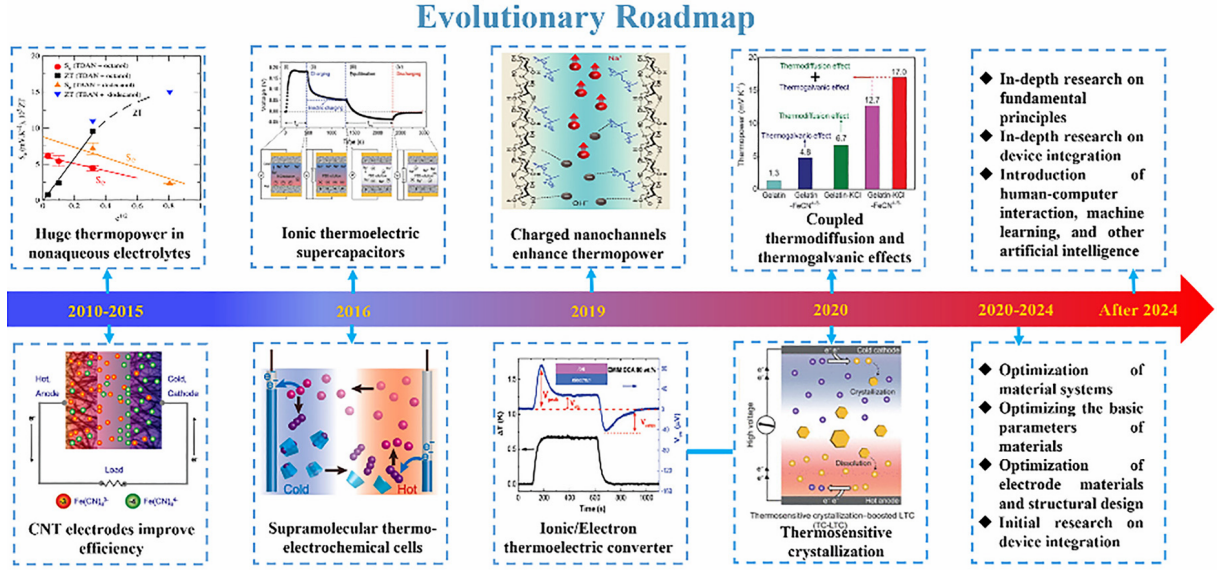


Figure 1.2: Representative evolutionary roadmap of ionic thermoelectrics, adapted from Shang et al. [2].

The roadmap in Figure 1.2 separates several stages of development. From 2010 to 2015, non-aqueous electrolytes and nanostructured electrodes showed that thermogalvanic systems could be improved by chemistry and interface engineering. Around 2016, thermally charged ionic capacitors clarified how ion redistribution can produce large thermovoltages. By 2019 and 2020, TD–TG-coupled systems and thermosensitive crystallization showed that high thermopower can be generated not only by choosing a redox couple, but also by constructing spatial gradients and selective molecular interactions. Recent reviews therefore frame TG development around redox thermodynamics, solvation-structure control, gel electrolyte design, device integration, and data-assisted materials design [2].

This development establishes a mechanism-based order for electrolyte thermoelectrics. TD provides the high-voltage case for ion redistribution, whereas TG defines a redox-entropy problem in which redox couples, solvation environments, additives, and gel matrices jointly control thermopower.

1.1.2 Thermodiffusion and Thermogalvanic Conversion

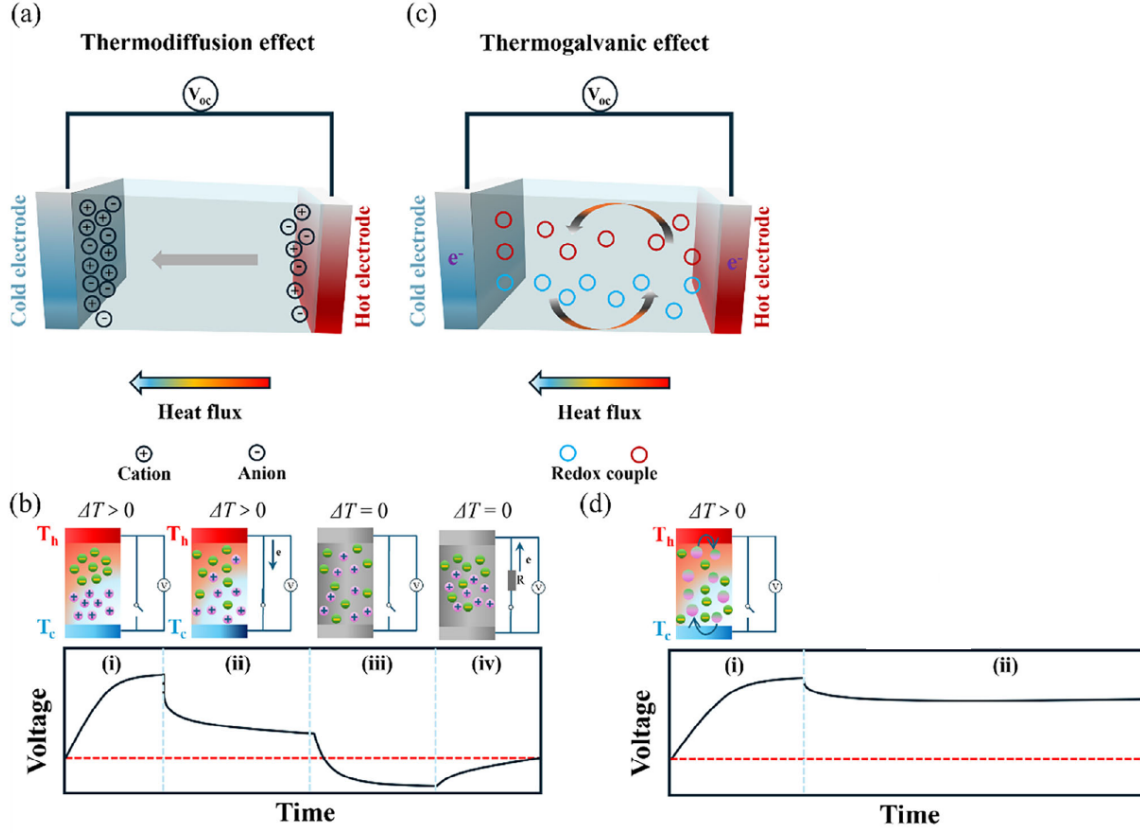


Figure 1.3: Mechanistic distinction between thermodiffusion and thermogalvanic effects in ionic thermoelectric devices, adapted from Lin et al. [3].

Thermodiffusion originates from the Soret effect. Under a temperature gradient, cations and anions can migrate with different thermodiffusion coefficients. This asymmetric ion redistribution generates concentration polarization and an electric double layer near blocking electrodes. Thermally charged capacitors use this effect through a cycle of voltage buildup, charging, equilibration, and discharge. They can produce very large apparent ionic Seebeck coefficients, but the practical output is commonly tied to charge-discharge operation and is therefore intermittent [8].

Thermogalvanic cells operate differently. A redox couple is present in the electrolyte, and the two identical electrodes are held at different temperatures. Because the redox potential depends on temperature, the hot and cold electrodes acquire different equilibrium potentials. Once the cell is connected to an external circuit, oxidation and reduction occur in opposite directions at the two electrodes, while redox ions migrate through the electrolyte and are regenerated. This reversible redox cycle allows continuous current generation as long as the temperature gradient, electrode kinetics, and redox mass transport are maintained [9].

Table 1.1: Practical distinction between thermodiffusion-dominated and thermogalvanic ionic thermoelectrics.

Aspect	Thermodiffusion-dominated systems	Thermogalvanic systems
Primary carrier process	Cation/anion migration under a temperature gradient and ion accumulation near blocking interfaces	Temperature-dependent redox reaction at electrodes, coupled with redox-ion diffusion in the electrolyte
Device expression	Thermally charged capacitors and ionic thermoelectric capacitors	Thermoelectrochemical cells, also called thermogalvanic cells
Main advantage	Very high ionic Seebeck coefficient is possible because ions have large Eastman entropy of transfer	Continuous output is possible because redox species are reversibly consumed and regenerated
Main limitation	Output often decays after connection to a load and requires a charging/discharging sequence	Voltage is lower than many thermodiffusion gels, and performance depends strongly on redox kinetics and mass transport
Representative design question	How to maximize asymmetric ion migration and preserve the ion gradient	How to enlarge redox entropy while preserving reversibility, solubility, and diffusion

The practical distinction is therefore not only the sign or magnitude of the Seebeck coefficient. It is the time structure of the output. A thermodiffusion device can produce a large open-circuit voltage because ion migration and electric-double-layer formation store charge at the interfaces. Once a load is connected, however, the stored charge can be discharged and the voltage may relax unless the system is reset or operated in a cycle. A thermogalvanic cell also develops concentration gradients during operation, but the sustained current is supplied by reversible redox reactions rather than by the discharge of a previously charged interface.

This difference matters when comparing reported thermopowers. A TD-dominated gel with S_i of tens of mV K^{-1} may be excellent for thermal sensing or thermally chargeable capacitors, but it is not automatically a continuous power source. A TG cell with a smaller S_{TG} can be more attractive for steady low-grade heat harvesting because the redox couple closes the ionic transport loop. Thermopower is therefore only the first filter; reversibility, mass transport, solvation stability, and gel architecture determine whether that thermopower can be converted into practical output.

1.2 Solvation Entropy Difference in Thermogalvanic Cells

For a thermogalvanic cell, the thermopower is the voltage produced per unit temperature difference:

$$S_{\text{TG}} = \frac{\Delta V}{\Delta T}.$$

For an ideal reversible redox process with n transferred electrons, the temperature coefficient of the electrode potential is linked to the entropy change of the redox reaction:

$$S_{\text{TG}} \approx \frac{\Delta S_{\text{redox}}}{nF},$$

where F is the Faraday constant. This expression is the simplest way to see why thermogalvanic electrolytes can exceed traditional electronic Seebeck coefficients. The voltage is not controlled only by carrier diffusion in a solid; it is controlled by how much entropy changes when a redox species changes oxidation state in a given solvent and ionic environment [10, 11].

In real electrolytes, this entropy is not an intrinsic property of the redox center alone. Oxidized and reduced species may have different charge density, coordination geometry, spin state, hydration shell, counterion association, and hydrogen-bonding environment. A solvent or additive that interacts more strongly with one redox state than the other changes the difference in local molecular order between the two states. This is the solvation entropy difference, and it has become one of the most important design variables for thermogalvanic electrolytes [11].

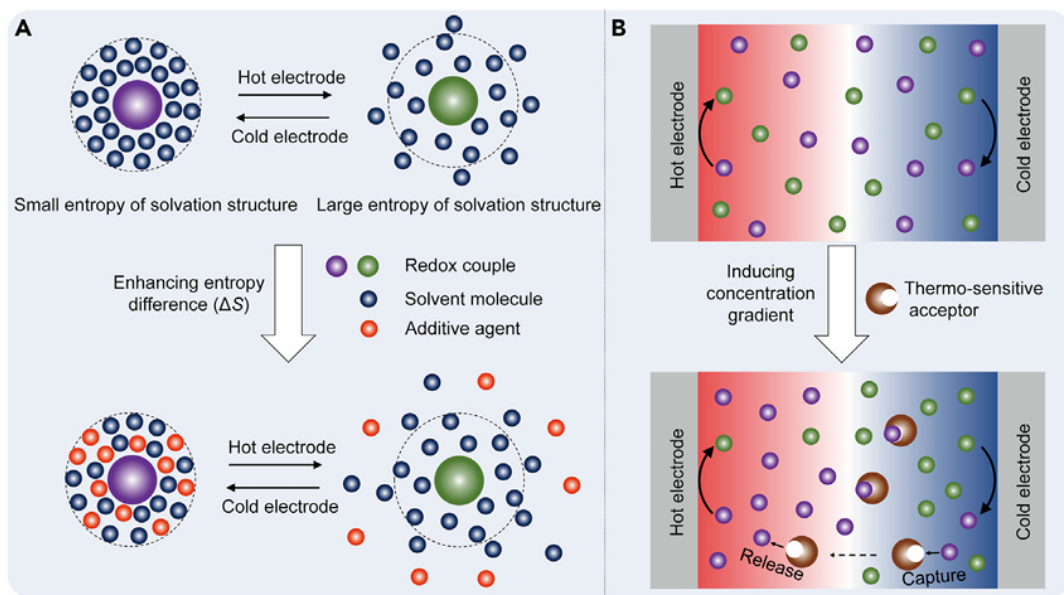


Figure 1.4: Two general routes to higher thermogalvanic thermopower: increasing the solvation-structural entropy difference and inducing a redox concentration gradient. Adapted from Duan et al. [1].

The second route is concentration asymmetry. The Nernst equation indicates that the electrode potential depends on the activity and concentration ratio of the oxidized and reduced species. If temperature-dependent solubility, crystallization, host-guest binding, or phase transition creates different redox ratios at the hot and cold sides, thermopower can be amplified beyond

the baseline redox entropy contribution. This strategy is powerful but difficult: concentration gradients are thermodynamically unstable and tend to relax by diffusion. Recent work therefore attempts either to regenerate the gradient continuously or to lock it into a gel, supramolecular host, or phase-separated structure [12–14].

1.2.1 Role of Solvation Entropy

Solvation entropy is central because redox in a liquid is never just electron transfer. When a species is oxidized or reduced, the solvent must reorganize to screen the new charge distribution. If the oxidized state orders nearby solvent molecules more strongly than the reduced state, the entropy difference between the two states becomes large. A large entropy difference makes the redox potential more temperature dependent, which directly increases S_{TG} .

This principle also explains why redox couples that look chemically similar can perform differently in different solvents. Water, methanol, acetone, DMSO, ethylene glycol, and deep eutectic solvents impose different hydrogen-bonding, polarity, donor-number, and viscosity environments. These properties can change the effective radius of the solvation shell, the activity coefficient of each redox species, and the tendency of one redox state to crystallize or bind to an additive. In this sense, the solvent is not a background medium; it is part of the thermogalvanic material.

The same logic applies in polymer gels. A hydrogel contains water, polymer functional groups, redox ions, counterions, and sometimes organic cosolvents or fillers. Each component can change the local entropy of the redox states. A polymer matrix that improves mechanical strength but makes both redox states equally immobile may not improve thermopower. A matrix that selectively stabilizes one redox state, induces a temperature-dependent concentration profile, or maintains a hydrated transport channel can change the observed thermogalvanic response.

1.2.2 Entropy Difference and Concentration Difference

It is helpful to distinguish two routes to larger thermopower. The first route is an intrinsic redox-entropy route: increase the difference in entropy between the oxidized and reduced states. This can be achieved by selecting a redox pair with different charge density, by using a ligand environment that changes coordination or spin state, or by using solvent/additive chemistry that selectively reorganizes one state. The methanol-water ferri/ferrocyanide example belongs to this route because methanol changes the solvation shell and raises the thermopower from the

aqueous baseline [10].

The second route is a concentration-gradient route. The redox potential depends on the ratio of oxidized and reduced species. If a thermosensitive host, crystal, gel, or phase transition changes this ratio differently at the two electrodes, the Nernst contribution increases the voltage. The guanidinium-induced crystallization of ferri/ferrocyanide is a representative example: temperature-dependent crystallization and dissolution sustain a redox concentration difference and increase thermopower to about 3.73 mV K^{-1} [13]. The host-guest binding of triiodide by α -cyclodextrin follows the same broad idea but uses supramolecular selectivity instead of crystallization [12].

These two routes are not independent in practice. A guanidinium salt can change hydration structure while also inducing crystallization. A polymer can change redox-ion diffusion while also creating a local concentration profile. A cosolvent can alter the entropy difference and the solubility difference at the same time. For this reason, high thermopower should be interpreted mechanistically rather than treated as a single number.

1.2.3 From Thermopower to Recoverable Device Output

Thermopower is an essential material descriptor, but it is not a sufficient measure of heat-harvesting performance. The open-circuit relation $V_{\text{oc}} = S\Delta T$ states the voltage available before current is drawn; it does not report the internal resistance, redox diffusion rate, electrode overpotential, heat leakage, or decay of the temperature gradient. Consequently, two cells with the same S_{TG} can deliver very different power and energy. The recent reviews converge on a hierarchical evaluation: thermopower identifies the voltage-generating chemistry; current-voltage and impedance measurements identify kinetic losses; and energy delivered over a defined operating period or cycle establishes whether the chemistry is useful at the device level [1, 2, 4, 5].

For a resistive cell operated near steady state, the maximum areal power density is commonly obtained from a load sweep, with the maximum occurring near load matching. Normalizing this value by $(\Delta T)^2$ removes the trivial quadratic dependence of electrical power on temperature difference, but it does not remove dependence on electrode area, electrode spacing, orientation, or thermal boundary conditions. Carnot-relative efficiency adds a direct heat-to-electricity criterion, yet it remains sensitive to the hot-side temperature and to how the heat input and effective thermal conductivity are defined. These metrics are therefore informative only when accompanied by cell geometry, electrode materials, temperature history, and the time at which the

measurement was taken [1].

The distinction between continuous TG cells and TD-dominated thermal charging is especially important here. A peak power extracted immediately after thermal charging can characterize a transient device, but it cannot establish sustained power generation. For cyclic TD devices, energy delivered per complete charge–discharge cycle and efficiency based on the corresponding thermal input are more defensible. For TG devices intended for continuous conversion, the relevant evidence is the stabilized current or energy delivered while a fixed temperature difference is maintained, together with long-duration voltage and power traces. Applying the conventional electronic $ZT = S^2\sigma T/\kappa$ without accounting for slow ionic redistribution and interfacial charging can therefore be misleading [4, 5].

Table 1.2: A mechanism-aware hierarchy for evaluating ionic thermoelectric materials and devices.

Evaluation level	Recommended quantity	Question answered	Main caveat
Material voltage	$S = \Delta V/\Delta T$ from multiple stabilized ΔT values	How strongly does the material or redox environment generate open-circuit voltage?	Does not quantify output under load
Transport and interface	Ionic/redox conductivity, diffusion coefficient, impedance, and charge-transfer resistance	Which process limits current and response time?	Values depend on electrode and geometry
Instantaneous device output	Full current–voltage curve and $P_{\max}/(\Delta T)^2$	What peak power can the stated cell deliver?	Can overstate a decaying or cyclic output
Operational yield	Stabilized power and time-integrated energy, or energy per complete cycle	How much usable electrical energy is recovered in the stated working mode?	Requires an explicit duration or cycle definition
Conversion efficiency	Electrical energy divided by measured thermal input, with Carnot-relative efficiency where appropriate	How effectively is heat converted under the specified boundary conditions?	Sensitive to heat-loss model and temperature definition

This hierarchy changes how material strategies should be judged. Increasing redox entropy is valuable only if the same intervention does not cause severe ion pairing, precipitation, viscosity growth, or slow electrode kinetics. Likewise, a gel that suppresses convection may preserve a larger temperature difference, but its net benefit depends on whether redox transport remains fast enough under load. The appropriate optimization target is thus recoverable energy under a declared operating mode, not the largest isolated thermopower.

1.3 Redox Couples

The redox couple determines the sign, magnitude, and reversibility of a TG response. Inorganic and metal-complex redox pairs are widely investigated because they offer reversible electrochemistry, high solubility, and relatively simple electrolyte preparation. Organic redox couples have attracted increasing attention because proton-coupled electron transfer and phase-transition chemistry can introduce additional entropy changes. The redox chemistry is therefore organized below according to the origin of the entropy difference, rather than by publication year.

1.3.1 Ferri/Ferrocyanide Redox Couple

The ferri/ferrocyanide couple, $[\text{Fe}(\text{CN})_6]^{3-/4-}$, is one of the most widely used p-type redox couples in aqueous TG cells. Its use is supported by reversible electrochemistry, good water solubility, and well-established electrode kinetics. Its aqueous thermopower is commonly around 1.4 mV K^{-1} , which is already large compared with electronic thermoelectric coefficients [9].

The continuing use of ferri/ferrocyanide also reflects its sensitivity to the surrounding electrolyte environment. Methanol changes the solvation shell and raises the thermopower. Guanidinium and urea perturb hydration shells. Guanidinium chloride can induce thermosensitive crystallization of the ferrocyanide-rich phase, creating a concentration difference. Gel matrices can immobilize the liquid while preserving redox diffusion. These variations make ferri/ferrocyanide a model system for examining how solvation, additives, and soft matrices affect redox entropy.

1.3.2 Iron Aqua Complexes and Counterion Sensitivity

The $\text{Fe}^{2+}/\text{Fe}^{3+}$ system has a simpler coordination picture than ferri/ferrocyanide, but it is more sensitive to pH, hydrolysis, and counterion chemistry. The oxidized and reduced aqua complexes have different charge density and hydration structures, so the entropy difference can be tuned by changing the anion and solvent. $\text{Fe}^{2+}/\text{Fe}^{3+}$ is therefore relevant for studying solvation effects, but it also makes the system more difficult to standardize across laboratories.

Solvent-donor-number studies illustrate this sensitivity. Solvents such as DMSO and tetramethylene sulfone change the first solvation shell around iron ions and therefore alter the thermopower. This type of work shifts the field from empirical additive screening toward molecular descriptors for solvation entropy [11].

1.3.3 Iodide/Triiodide and Supramolecular Selectivity

The iodide/triiodide couple is frequently treated as an n-type counterpart to ferri/ferrocyanide. Its baseline thermopower is lower than many optimized ferri/ferrocyanide systems, but its chemistry is unusually responsive to selective binding. Iodide is comparatively hydrophilic, whereas triiodide is more polarizable and can interact with hydrophobic cavities or polymer domains. This difference makes the system suitable for host-guest regulation.

The α -cyclodextrin case illustrates the principle. Cyclodextrin can selectively bind triiodide and create a temperature-dependent concentration asymmetry. The thermopower increases from the baseline iodide/triiodide value to about 1.97 mV K^{-1} [12]. The mechanism is notable because it does not rely on changing the redox couple itself. It uses supramolecular chemistry to change the distribution of one redox state.

1.3.4 Metal-Complex and Organic Redox Couples

Metal-complex redox couples such as cobalt bipyridyl complexes are attractive because ligands can tune charge distribution, spin state, solubility, and electrochemical reversibility. Their main limitation is transport: ionic liquid and deep eutectic solvent environments can give stable operation, but high viscosity often lowers diffusion and power density. The choline chloride-ethylene glycol deep eutectic solvent example shows that solvent chemistry can expand the usable temperature window while keeping a measurable thermogalvanic response [15].

Organic redox couples are a newer direction. Quinone/hydroquinone derivatives, methyl viologen, and pyrazine-based molecules are attractive because molecular structure can be deliberately tuned. Some organic redox systems also involve proton-coupled electron transfer, where electron transfer is coupled to proton release or uptake. This coupling can create large entropy changes, especially near a pH or phase transition. The hydrogel-nanoparticle thermocell based on volume phase transition is an example of using a soft phase transition to amplify the redox thermopower [14].

Table 1.3: Representative redox couples and reported thermopowers. Each row uses one main source for the specific value or example.

Redox couple	Representative thermopower	Main chemical origin or design logic	Main source
$[\text{Fe}(\text{CN})_6]^{4-/3-}$ in water	about 1.4 mV K^{-1}	Stable outer-sphere redox pair; large charge difference and strong hydration make it a widely used p-type aqueous system	[9]
$[\text{Fe}(\text{CN})_6]^{4-/3-}$ with methanol-water solvent	about 2.9 mV K^{-1}	Selective solvation-shell rearrangement of the reduced species increases redox solvation entropy difference	[10]
$[\text{Fe}(\text{CN})_6]^{4-/3-}$ with guanidinium/urea	about 4.2 mV K^{-1}	Chaotropic and hydrogen-bonding interactions reorganize hydration structures around ferri/ferrocyanide	[9]
$[\text{Fe}(\text{CN})_6]^{4-/3-}$ with thermosensitive crystallization	about 3.73 mV K^{-1}	Guanidinium-induced crystallization and dissolution create a redox concentration difference between electrodes	[13]
I^-/I_3^- with α -cyclodextrin	about 1.97 mV K^{-1}	Host-guest binding of triiodide creates a temperature-dependent concentration gradient	[12]
$\text{Fe}^{2+/3+}$ in solvent-engineered electrolytes	up to several mV K^{-1} depending on solvent	Solvent donor number and hydration-shell structure tune the entropy difference between ferric and ferrous states	[11]

Redox couple	Representative thermopower	Main chemical origin or design logic	Main source
[Co(bpy) ₃] ^{2+/3+} in deep eutectic solvent	about 1.67 mV K ⁻¹	Coordination redox chemistry in a hydrogen-bond-rich solvent environment	[15]
Quinone-based organic systems	about 6.7 mV K ⁻¹ in a phase-transition thermocell	Proton-coupled electron transfer and hydrogel nanoparticle phase transition create a large entropy response	[14]
Pyrazine-based organic redox species	above 3 mV K ⁻¹	Molecular redox structure can be tuned without relying on metal complexes	[16]

Table 2 indicates a shift in design emphasis. Early work mainly treated the redox couple as a fixed electrolyte component. Recent studies increasingly consider the redox couple and its environment together. Ferri/ferrocyanide, for example, has been used to examine counterion binding, methanol-water solvation, guanidinium-induced crystallization, polymer confinement, and gel mechanics.

1.4 Solvents and Additives

The redox entropy in a TG electrolyte is determined by the redox species and the molecular environment that reorganizes during electron transfer. Two levels of environmental control should be distinguished. Solvent engineering changes the continuous liquid phase in which the redox couple is dissolved. It therefore acts broadly on dielectric screening, donor and acceptor ability, hydrogen-bond structure, viscosity, redox solubility, and electrochemical stability. Additive engineering introduces a secondary component selected for a more specific interaction, such as preferential binding, chaotropic hydration-shell disruption, counterion pairing, thermosensitive crystallization, host-guest recognition, or polymer-chain association.

The distinction is mechanistic rather than only compositional. A water-alcohol mixture is treated as solvent engineering when the added alcohol changes the bulk solvation environment and transport properties. Guanidinium salts, cyclodextrins, betaine, or crosslinking agents are

treated as additives when their main role is selective interaction with one redox state, one counterion, or the gel network. Mixed cases exist, especially in deep eutectic solvents, high-loading cosolvent systems, and hydrogels, but the classification should be based on the dominant thermodynamic role rather than the ingredient name.

Thermopower can be increased either by increasing the solvation-structural entropy difference, ΔS , or by establishing a redox concentration-ratio difference, ΔC_r . This distinction provides a mechanistic basis for grouping the later literature. Methanol-water ferri/ferrocyanide electrolytes are a mainly solvation-entropy example [10]. Guanidinium-induced crystallization is a concentration-ratio example. Cyclodextrin-triiodide chemistry is a host-guest concentration-ratio example [12, 13]. Donor-number tuning of $\text{Fe}^{2+/3+}$ is a first-solvation-shell example [11]. Deep eutectic solvents and ionic liquids are broader solvent-system examples, because they change volatility, electrochemical window, ion association, and diffusion at the same time.

From the application side, solvent design also determines whether a TG cell can operate beyond a narrow laboratory window. Water gives high ion mobility and simple electrochemistry, but it evaporates, freezes, boils, and can trigger side reactions at high voltage. Organic cosolvents and cryoprotectants can extend low-temperature operation, but they may increase viscosity or reduce biocompatibility. Ionic liquids and deep eutectic solvents reduce volatility and broaden temperature range, but they often introduce stronger ion-ion correlation and slower diffusion. Thus, solvent engineering is not a simple route to a larger S_{TG} ; it is a four-way compromise among thermopower, solubility, conductivity, and stability.

Table 3 separates bulk-solvent strategies from additive strategies. The former modify the continuous electrolyte environment, whereas the latter rely on a specific molecular or ionic interaction superimposed on that environment. This separation is necessary because similar thermopower values can arise from different physical origins: solvent-shell reorganization, temperature-dependent solubility, selective binding, crystallization, or polymer-network effects.

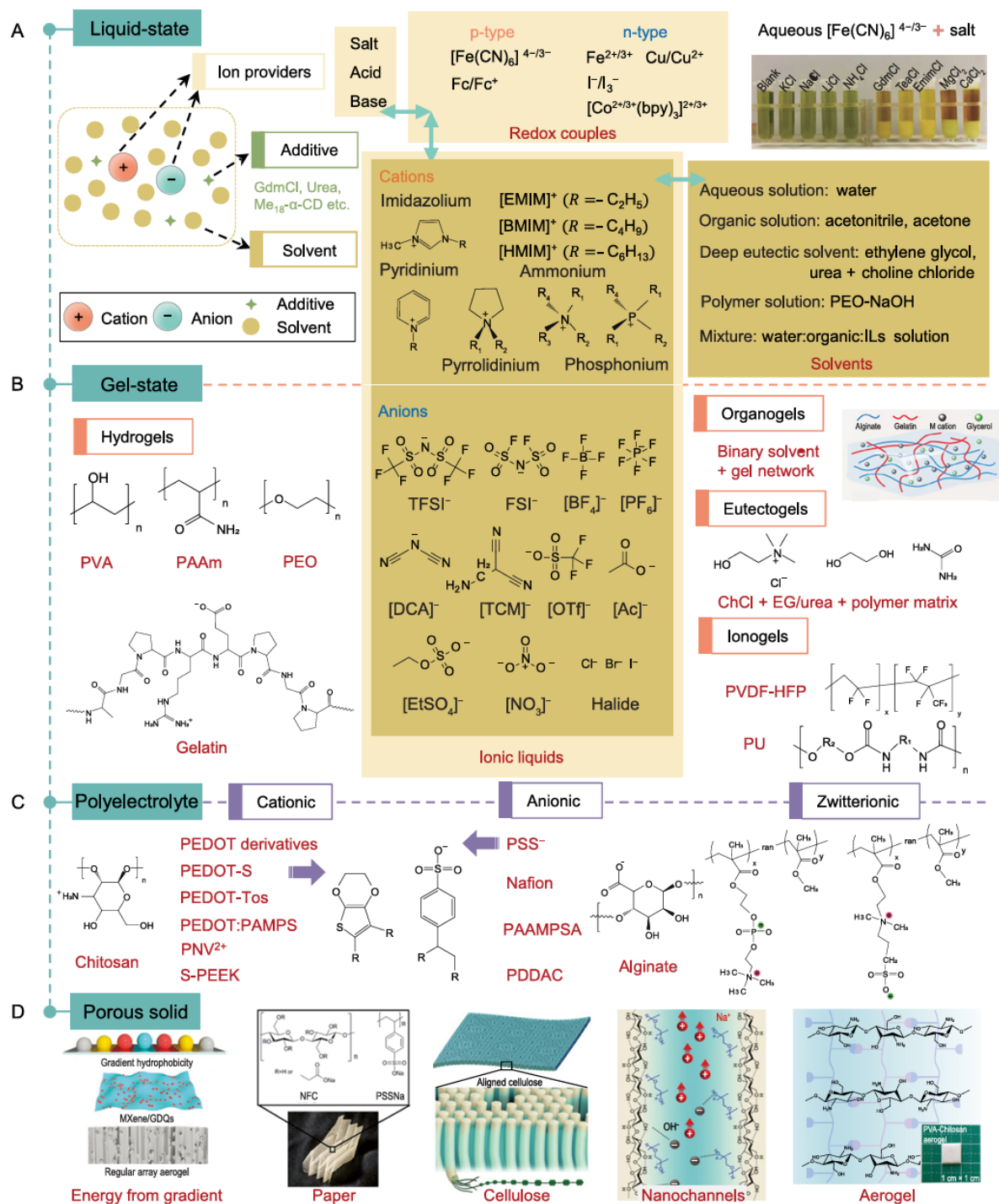


Figure 1.5: Material classes and representative constituents of ionic thermoelectrics, spanning liquid-state electrolytes, gel-state electrolytes, polyelectrolytes, and porous solids. Adapted from Xu et al. [4].

Table 1.4: Representative solvent and additive strategies. Solvent rows refer to changes in the bulk liquid phase; additive rows refer to selective molecular, ionic, or polymer-network interactions.

Redox pair	Component	Main role	Source
$[\text{Fe}(\text{CN})_6]^{4-/3-}$	Solvent: methanol-water mixture	Rearranges the solvation shell and increases the redox solvation entropy difference; the reported thermopower rises to about 2.9 mV K^{-1}	[10]
$[\text{Fe}(\text{CN})_6]^{4-/3-}$	Additives: guanidinium and urea	Perturbs hydration shells through chaotropic and hydrogen-bonding effects; the reported thermopower reaches about 4.2 mV K^{-1}	[9]
$[\text{Fe}(\text{CN})_6]^{4-/3-}$	Additive: guanidinium chloride	Induces thermosensitive crystallization/dissolution of $[\text{Fe}(\text{CN})_6]^{4-}$ and creates a redox concentration difference	[13]
I^-/I_3^-	Additive: α -cyclodextrin	Selectively binds triiodide and establishes a temperature-dependent concentration gradient	[12]
$\text{Fe}^{2+/3+}$	Solvents with different donor numbers	Changes the first solvation shell and the entropy difference between Fe^{3+} and Fe^{2+}	[11]
$[\text{Co}(\text{bpy})_3]^{2+/3+}$	Solvent: ChCl-EG deep eutectic solvent	Provides a low-volatility hydrogen-bonding solvent network for thermogalvanic operation at middle temperature	[15]

Redox pair	Component	Main role	Source
Quinone redox system	Gel NPs	Uses volume phase transition and pH-sensitive PCET chemistry to amplify the thermal response	[14]

The common theme is that solvent and additive design must be judged by two entropy terms rather than by a single ingredient name. One term is the solvation-structural entropy difference between redox states. The other is the concentration-ratio contribution that appears when the oxidized and reduced species are not distributed identically at the hot and cold electrodes. The most successful designs often use both: guanidinium chemistry modifies ferri/ferrocyanide solvation while also enabling thermosensitive crystallization, and cyclodextrin chemistry increases the effective concentration asymmetry of triiodide through selective host-guest binding.

1.4.1 Solvent Engineering

Solvent engineering acts on three properties at once. The first is redox entropy, because the solvent shell reorganizes during electron transfer. The second is solubility, because the oxidized and reduced species may have different temperature-dependent solubilities. The third is transport, because viscosity controls how fast redox ions can diffuse between the electrodes. Good solvent design therefore requires a compromise: the solvent must make the redox states entropically different, but it must not immobilize the ions so strongly that current output collapses.

Methanol-water ferri/ferrocyanide electrolytes illustrate a relatively clean solvation effect. The redox couple is unchanged, but the local solvent environment changes enough to increase thermopower. This provides a clear mechanistic control: thermogalvanic thermopower can be modified without introducing a new redox molecule [10].

The methanol case also clarifies what is meant by "solvent engineering." The added alcohol does not simply dilute the aqueous electrolyte. It changes the relative organization of solvent molecules around the two charge states of the redox pair. Because $[\text{Fe}(\text{CN})_6]^{4-}$ has a larger charge magnitude than $[\text{Fe}(\text{CN})_6]^{3-}$, it is more sensitive to changes in the hydrogen-bonding environment. The thermopower increase from about 1.4 to 2.9 mV K⁻¹ therefore comes from making the solvation entropy difference between the reduced and oxidized species larger. The mechanistic implication is that a widely used redox couple can be treated as a solvent-responsive thermodynamic unit rather than as a fixed electrolyte component.

Organic cosolvents can also work through solubility and temperature-dependent phase behavior rather than through a purely local solvation shell. A bacterial-cellulose hydrogel containing ferri/ferrocyanide in a water/propylene glycol medium, where propylene glycol promotes gradual crystallization of $K_4[Fe(CN)_6]$ and thereby introduces a redox concentration contribution to thermopower [17]. In that case, spectroscopic evidence suggested that the cyano ligand environment did not change strongly; the performance increase was instead assigned mainly to the concentration-ratio term. Ethylene glycol in PAAm-based ferri/ferrocyanide hydrogels provides another example of solvent design serving a device-level function: it suppresses freezing by disturbing the water hydrogen-bond network, while the different solubility of ferricyanide and ferrocyanide helps generate a thermally dependent redox distribution [18]. These cases show why "cosolvent" should not be treated as a single mechanism. It may tune solvation entropy, crystallization, freezing resistance, or several of these simultaneously.

For $Fe^{2+/3+}$ electrolytes, donor-number control provides a more direct connection between solvent coordination and thermopower. Solvents with different donor numbers can stabilize Fe^{3+} and Fe^{2+} to different degrees, changing the structure and disorder of the first solvation shell. The PAAm/water $Fe^{2+/3+}$ system shows that a high-donor-number solvent reduced the ionic Seebeck coefficient, whereas a lower-donor-number solvent raised it to about 2.49 mV K^{-1} [11]. Free-energy perturbation and molecular-dynamics calculations for water/acetone systems, where high acetone fraction was associated with a more ordered solvation shell and larger thermopower [11]. These studies matter because they move the field beyond empirical solvent screening: the target becomes the relative solvation entropy of the two redox states, which can in principle be predicted.

Deep eutectic solvents show a different balance. They are less volatile than water-rich electrolytes and can operate over broader temperature windows, but their viscosity can restrict ion diffusion. Their value is therefore not only high thermopower; it is the ability to combine redox entropy modulation with practical stability. This is especially relevant for devices that must operate outside a narrow ambient-temperature window [15].

The IL and DES literature should be read in this practical context. Ionic liquids provide non-volatile and thermally stable media when water evaporation or boiling would restrict operation, although their higher viscosity can reduce redox diffusion and power output [19]. A representative DES example is the $[Co(bpy)_3]^{2+/3+}$ thermocell in choline chloride/ethylene glycol, where a hydrogen-bond-rich, low-volatility solvent network supported middle-temperature operation with a thermopower of about 1.67 mV K^{-1} [15]. Water-containing DES gels then add another layer: the DES adjusts redox solvation and freezing resistance, while the gel network supplies

dimensional stability.

Counterion and ligand effects are closely related to solvent effects, because they determine how tightly the redox center is paired or coordinated. In $\text{Fe}^{2+/3+}$ systems, weakly coordinating anions such as perchlorate can reduce ion pairing and stabilize aquo complexes, producing higher thermopower and better effective conductivity than more strongly interacting counterions [9]. Ligand design plays an analogous role in metal-complex redox couples. Introducing ethylenediamine ligands into copper complexes, for example, changes the coordination environment and reaction entropy, illustrating that the solvent shell and the primary coordination sphere should be considered together. These cases sit between "redox-couple selection" and "solvent selection": the redox formula changes, but the performance gain is still largely environmental and entropic.

1.4.2 Additives as Local Entropy Modifiers

Additives are attractive because they can be introduced at low concentration and targeted toward one interaction. Guanidinium chemistry is the clearest example in ferri/ferrocyanide systems. In one case, guanidinium and urea change the hydration shell and increase the redox solvation entropy difference [9]. In another case, guanidinium chloride induces thermosensitive crystallization, so the key effect is a concentration difference rather than only a solvation-shell difference [13]. These two cases should not be merged into a single "guanidinium additive" mechanism. They use related chemistry, but the dominant thermodynamic contribution is different.

The guanidinium/urea aqueous ferri/ferrocyanide work is best understood as a hydration-shell design study. Guanidinium is a chaotropic cation with strong interactions toward the highly charged ferrocyanide ion, while urea reorganizes the surrounding hydrogen-bond network. Together they amplify the difference between the solvation environments of $[\text{Fe}(\text{CN})_6]^{4-}$ and $[\text{Fe}(\text{CN})_6]^{3-}$, raising the thermopower to about 4.2 mV K^{-1} [9]. The work made the solvation-entropy mechanism experimentally tangible: the additive was not introduced as an inert salt, but as a molecular tool for making one redox state more structurally distinct from the other.

The thermosensitive-crystallization work using guanidinium chloride should be kept separate. Here the major design move is the selective formation and dissolution of ferrocyanide-associated crystals at different temperatures. The cold side becomes depleted or enriched in one redox species relative to the other, so the Nernst concentration term contributes to the open-circuit voltage. The reported thermopower of about 3.7 mV K^{-1} and high Carnot-relative efficiency come from a coupled improvement: concentration asymmetry boosts voltage, crystal

formation suppresses convection, and the local redox ratio can favor electrode reactions [13]. The result shows that a concentration gradient, usually assumed to relax away, can be stabilized by temperature-dependent association.

Cyclodextrin-based additives work through a third mechanism: supramolecular recognition. Instead of perturbing all hydration shells, cyclodextrin selectively binds triiodide. The binding changes the free concentration of one redox species, and the temperature dependence of that binding creates a redox concentration gradient. This mechanism explains why host-guest chemistry has become part of thermogalvanic design [12].

The same logic appears in later thermoresponsive iodine systems. PNIPAM- and methyl-cellulose-containing electrolytes can interact selectively with triiodide when the polymer becomes more hydrophobic above its transition temperature, changing the local concentration of active iodine species and even allowing apparent p/n conversion in some temperature regimes [20]. These systems are not merely "iodide redox couples in gels." They use the redox equilibrium, host or polymer affinity, and thermal phase behavior as a single coupled design element. Their advantage is strong thermal selectivity. Their limitation is that the operating temperature window is defined by the host or polymer transition, so response reversibility and cycling stability become as important as the absolute thermopower.

Some additives are introduced directly into hydrogels, where their function is both thermodynamic and mechanical. In a gelatin/ferri-ferrocyanide hydrogel, betaine acts as a zwitterionic additive that interacts with polymer chains while selectively rearranging the solvation structure of ferricyanide, giving a thermopower near 2.2 mV K^{-1} [21]. In another gel example, glutaraldehyde crosslinking in gelatin-based systems strengthens the matrix while also interacting with ferri/ferrocyanide, leading to a much larger reported ionic thermopower in a coupled ionic thermoelectric material. These cases show why hydrogel additives are harder to classify than liquid additives: the same molecule may tune polymer chain mobility, water retention, redox solvation, and mechanical integrity.

Filler additives add another layer of complexity. MXene-containing hydrogels can improve mechanical strength, self-healing behavior, and interfacial contact. However, conductive or redox-active fillers also bring risks. They may change electron-transfer pathways, alter the local temperature field, or make it harder to separate thermogalvanic voltage from other ionic or electronic effects. For this reason, filler-based TG hydrogels should report mechanical durability and power output together with controls that isolate the redox contribution.

1.4.3 Transport and Stability Trade-Offs

Solvents and additives can improve thermopower while damaging other parts of the device. A strong interaction can increase entropy contrast but also slow diffusion. Crystallization can build a concentration difference but can also cause sedimentation, orientation dependence, or long-term instability. Organic solvents can suppress freezing but may reduce biocompatibility or change electrode compatibility. Deep eutectic solvents can reduce volatility but may increase viscosity. Therefore, an informative additive strategy must report not only S_{TG} , but also conductivity, stability, reversibility, and the conditions under which the additive remains chemically benign.

Across the reviews, the most reliable design rule is therefore not "add a chaotrope" or "use an organic solvent." It is to identify the dominant thermodynamic lever and then check the transport penalty. If the lever is solvation entropy, the evidence should include changes in redox solvation, ion-pairing, or coordination structure. If the lever is concentration asymmetry, the evidence should show selective binding, crystallization, or phase partitioning under a temperature gradient. If the lever is a polymer-assisted effect, the evidence should distinguish redox entropy from thermodiffusion and from mechanical improvements. Without this distinction, different works can appear to be variations of the same additive recipe even when the dominant thermodynamic contribution is different.

1.5 Hydrogel Matrices

Gel matrices address the most immediate weakness of liquid thermogalvanic cells: leakage and difficult packaging. In a gel thermocell, the matrix retains water or organic solvent, immobilizes the electrolyte shape, improves contact with soft or curved surfaces, and allows wearable or flexible device formats. However, a gel is not only a container. Polymer functional groups can interact with redox ions, counterions, and solvent molecules, thereby changing solvation, diffusion, crystallization, and mechanical stability at the same time [17, 21, 22].

Gelatin illustrates the possibility of coupling thermodiffusion and thermogalvanic contributions. A gelatin-KCl-ferri/ferrocyanide system reached a thermopower of 17 mV K^{-1} , far above the baseline thermogalvanic coefficient of ferri/ferrocyanide, because the charged gelatin network and KCl thermodiffusion contributed strongly in addition to the redox reaction [8]. This system should not be confused with a purely thermogalvanic record; it is a coupled ionic thermoelectric system.

Poly(vinyl alcohol) (PVA), polyacrylamide (PAAm), bacterial cellulose, and double-network hydrogels represent the main polymer directions in thermogalvanic gels. PVA offers simple processing, hydroxyl-rich hydration, and mechanically trainable networks. PAAm provides high water content and can be combined with ionic or covalent crosslinks for toughness. Bacterial cellulose offers aligned nanofibrils and mechanical reinforcement. Composite gels add cellulose nanofibers, MXene, carbon materials, or supramolecular components to balance thermopower, ion transport, and deformation resistance [22–25].

The design logic of gel thermogalvanic cells is therefore multiscale. At the molecular scale, the matrix should enhance redox solvation contrast or concentration asymmetry. At the transport scale, it should preserve redox-ion diffusion and electrode access. At the device scale, it should maintain hydration, temperature difference, and mechanical contact during bending, drying, or repeated thermal cycling. This multiscale requirement explains why recent reviews treat polymer matrices, solvent composition, additives, and device architecture as coupled design variables rather than separate optimization steps [2, 4, 6].

The historical development of gel TG electrolytes can be divided into three stages. The first stage was leakage prevention: polymers were used mainly to hold liquid electrolyte in place. The second stage was compatibility screening: researchers asked which polymers could gel ferri/ferrocyanide or metal-complex electrolytes without suppressing redox diffusion. The third stage is active polymer design: the matrix is expected to tune solvation, promote selective ion transport, regulate crystallization, and survive mechanical deformation. Representative early studies established gel compatibility and leakage-free operation [19, 26, 27]; more recent work treats polymer chemistry as an active handle for ion dynamics, mechanics, and heat harvesting [18, 21, 22].

Early ferri/ferrocyanide gels reveal why polymer compatibility cannot be assumed. Poly(sodium acrylate) gels could maintain voltammetric behavior close to aqueous ferri/ferrocyanide and pass simple inversion tests, suggesting that the anionic redox species remained mobile in a quasi-solid matrix [26]. PVA was attractive because of its processability and mechanical strength, but high ferri/ferrocyanide loading could cause flocculation or reduced output in some formulations [19]. CMC and PAAm were later identified as effective hosts for higher-concentration ferri/ferrocyanide gels, whereas some polyurethane or physically gelled systems showed poor power output because redox transport was restricted [19]. This early screening literature is not obsolete. It establishes a basic rule that remains valid in modern hydrogel designs: a good matrix must be chemically compatible with the redox couple before it can be used for more ambitious solvation or mechanical engineering.

Cellulose-based quasi-solid electrolytes introduced a different design idea. Instead of dissolving the polymer into the electrolyte, an insoluble cellulose network can lock the redox solution inside a porous skeleton. Cellulose-ferri/ferrocyanide systems show that the network provided mechanical reinforcement and a tortuous transport pathway, with an optimized cellulose content balancing redox diffusion, mechanical integrity, and thermal management [27]. Later bacterial-cellulose and cellulose-nanofiber systems extend this idea by using nanofibrils not simply as passive scaffolds, but as structural elements that can form aligned pathways, reinforce double networks, or cooperate with cosolvents and salts [17, 22]. The design lesson is that a polymer matrix can control three transport processes at once: ion diffusion, water/solvent retention, and heat leakage.

1.5.1 Gelatin Systems

Gelatin demonstrates that biopolymer networks can contribute actively to ionic thermoelectric behavior. The gelatin-KCl-ferri/ferrocyanide system is often cited because it reached a very high thermopower of 17 mV K^{-1} near room temperature [8]. The mechanism is not purely thermogalvanic. The redox couple contributes a TG voltage, while KCl thermodiffusion and the charged gelatin network contribute additional ionic thermopower. Gelatin is therefore a representative case for coupled TD-TG systems and shows why mechanism separation is required.

For thermogalvanic hydrogels, gelatin offers water retention, biocompatibility, and functional groups that can interact with ions. Its weakness is that mechanical stability and dehydration resistance may be insufficient without crosslinking, additives, or encapsulation. Gelatin-based systems are therefore best understood as a starting point for bio-derived soft thermogalvanic materials rather than as a universal matrix.

The later gelatin literature uses this starting point in two different ways. One route treats gelatin as a soft, ion-containing host for coupled thermodiffusion and thermogalvanic conversion. In these systems, a large reported ionic thermopower can originate from both the redox entropy and the Soret response of mobile ions, so the mechanism has to be decomposed before comparing it with a purely TG electrolyte. The other route treats gelatin as a chemically functional scaffold. Zwitterionic or crosslinking additives can change chain packing and water structure while also influencing redox solvation. Gelatin systems can couple crosslinking and ferri/ferrocyanide interactions with mechanical reinforcement and high thermopower. For a TG-focused literature review, the central point is not the magnitude alone, but that biopolymer-rich gels force the field to separate redox entropy, ion thermodiffusion, water management, and

network mechanics.

1.5.2 PVA and PAAm Systems

PVA and PAAm are widely used because their polymer networks are easy to tune. PVA has hydroxyl groups that help retain water and form physically crosslinked structures. Mechanical training or freeze-thaw processing can align polymer domains, producing ion-transport pathways. PAAm has high water content and can be combined with ionic crosslinks or double networks to improve toughness.

The central question for PVA and PAAm is how to improve mechanics without blocking redox diffusion. A stiff network may suppress leakage and improve wearability, but it can increase mass-transport resistance. A loose network may give high ion mobility, but it may dry, deform, or leak. Thermogalvanic hydrogel design therefore differs from ordinary hydrogel mechanics: the matrix must remain mechanically robust while keeping the redox couple electrochemically available.

PVA systems show why processing can be as important as composition. Hydroxyl groups help retain water and form reversible hydrogen bonds, while freeze-thaw or stretching treatments can create crystalline or aligned domains. PVA-based gel strategies can also connect processing to effective conductivity: salting-out or directional-freezing methods can create water-rich channels and anisotropic structures that reduce transport resistance. These examples point to a mesoscale design principle. If the polymer-rich phase carries mechanical load and the water-rich phase carries redox ions, the gel can be tough and conductive at the same time.

PAAm systems are relevant because the neutral amide network can accommodate aqueous redox species and can be combined with other networks. Early wearable TGC studies found PAAm to be one of the more effective ferri/ferrocyanide hosts when compared with several other gelators [19]. Later work uses PAAm as a structural framework for much more complex electrolytes. PAAm/PVA/cellulose-nanofiber triple-network hydrogels use PAAm for shape construction, PVA for flexibility, and cellulose nanofibers reinforce the network; the optimized structure reached an ionic conductivity of 168 mS cm^{-1} and a reported ZT of 0.06 [22]. Conductivity is therefore treated as a structural property of the gel, not only as a concentration property of the dissolved redox couple.

PAAm also appears in n-type $\text{Fe}^{2+/3+}$ gel systems, where the challenge is different. $\text{Fe}^{2+/3+}$ can be soluble and chemically robust, but its thermopower depends strongly on counterion, solvent, and coordination environment. Non-aqueous or mixed-solvent PAAm gels can com-

bine $\text{Fe}^{2+/3+}$ redox chemistry with anti-freezing behavior [18]. These systems matter for device integration because p-type ferri/ferrocyanide cells alone do not provide the ideal p/n module architecture. High-performance n-type gels remain a bottleneck for compact series-connected modules.

1.5.3 Cellulose and Composite Gels

Cellulose-based gels add another design route. Bacterial cellulose can provide nanofiber reinforcement and internal channels. Cellulose nanofibers can be combined with synthetic networks to improve toughness and dimensional stability. Composite gels can also include MXene, carbon materials, or conductive polymers, but these fillers must be used carefully. They can improve mechanical properties or electrode contact, yet they can also change interfacial reactions or introduce parasitic pathways.

The thermogalvanic hydrogel literature therefore increasingly emphasizes multiscale design. At the molecular scale, polymer functional groups and cosolvents tune solvation entropy. At the mesoscale, phase separation, nanofibers, and aligned channels control ion transport. At the device scale, encapsulation, electrode contact, and module geometry determine whether the temperature gradient and redox flux are maintained. A hydrogel is successful only when these scales are compatible.

Bacterial cellulose is especially attractive because it can combine mechanical strength with aqueous channels. In water/propylene glycol ferri/ferrocyanide systems, bacterial cellulose acts as a hydrogel scaffold while the cosolvent controls crystallization and low-temperature stability [17]. In hybrid cellulose-PAAm systems, cellulose nanofibers add reinforcement and can help organize ion-transport pathways [22]. A large thermopower does not lead to high power output if the redox species cannot move fast enough to support current under load.

Composite fillers such as MXene, carbon materials, and conductive polymers must be interpreted carefully. In a mechanical sense, they can dissipate energy, increase stretchability, and improve contact with electrodes. In an electrical or electrochemical sense, they may introduce additional charge-transfer sites or mixed ion-electron pathways. MXene-containing hydrogels can retain thermopower under repeated deformation, demonstrating the device value of robust soft materials. However, for mechanistic clarity, such systems should be reported with controls that separate matrix mechanics, electrode kinetics, and bulk TG voltage.

1.5.4 Thermoresponsive and Functional Polymer Matrices

Thermoresponsive polymers move beyond passive gelation by making the matrix itself temperature-selective. In iodine-based systems, PNIPAM or methyl cellulose can change hydrophobicity near a transition temperature and thereby alter the local activity of I_3^- [20]. This is conceptually similar to cyclodextrin binding, but the host is now a polymer network whose affinity changes with temperature. The benefit is strong thermal amplification and possible sign reversal. The drawback is that performance depends on the phase-transition window and on whether repeated heating and cooling preserve the same polymer morphology.

Functional side chains provide another route. Copolymer and zwitterionic designs use cationic or sulfobetaine groups that selectively interact with Fe^{3+} or ferri/ferrocyanide ions, changing hydration and redox entropy [21]. This line of work is relevant because it connects TG electrolytes with polymer physics: side-chain charge, segmental mobility, hydration, and ion association are all variables that can be designed. In a liquid electrolyte, the solvent shell is determined mainly by small molecules. In a functional polymer electrolyte, the redox species experiences a heterogeneous environment containing water, counterions, polymer segments, and sometimes nanoscale phase separation.

Chitosan-based and guanidinium-containing polymer networks illustrate how polymer chemistry can combine entropy and concentration effects. A chitosan/guanidine/melamine double-network $FeCl_3/FeCl_2$ system uses cationic groups to accelerate chloride migration and melamine to complex Fe^{3+} near the cold side, increasing both entropy and concentration-difference contributions [28]. The design extends beyond gelation. It uses polymer functional groups, redox-ion coordination, and thermodiffusive ion movement together. Such systems are powerful, but they also make mechanism attribution more demanding because TG, TD, and complexation-driven concentration gradients can all contribute to the measured voltage.

1.5.5 Gel Transport, Thermal Conductivity, and Device Stability

Gelation suppresses natural convection, which is one reason gels can improve effective thermal management relative to liquids. Joule 2021 emphasized that effective thermal conductivity in liquid thermocells includes convection under a temperature gradient, and that gel-based electrolytes can reduce this contribution [1]. This is a major advantage for maintaining a usable ΔT across the electrodes. The penalty is that gels often reduce redox diffusion. Therefore, modern hydrogel design tries to decouple thermal and ionic transport: suppress convective heat flow while preserving continuous aqueous or solvent-rich ion pathways.

Several strategies address this trade-off. Directional freezing can create aligned channels. Salting-out can form polymer-rich and water-rich domains. Nanofibers can reinforce the network without filling all transport space. Low-eutectic or mixed-solvent systems can keep water mobile over a wider temperature range. These are effective-conductivity strategies rather than purely mechanical modifications. The relevant conductivity for TG power output is not only the small-ion conductivity measured in a homogeneous electrolyte, but the combined resistance of redox diffusion, ion migration, polymer confinement, and electrode reactions.

Device stability adds another set of constraints. Wearable TGCs bend, stretch, dry, warm, cool, and repeatedly lose and regain contact with the skin or heat source. Flexible electrodes, gel electrolytes, and stable hydrogel-electrode interfaces must therefore be treated as parts of the same device problem. Strong and self-healing hydrogels extend the operating lifetime, but mechanical durability is only meaningful if the redox couple remains reversible and the thermopower is retained after deformation. Reports that include stretching cycles, cutting/healing cycles, anti-freezing tests, and long-term thermal cycling are therefore more informative than reports that only give an initial S_{TG} .

Table 1.5: Representative gel-matrix design principles for thermogalvanic electrolytes.

Matrix type	Typical redox/electrolyte choice	Main design function	Main source
Gelatin	KCl and $[\text{Fe}(\text{CN})_6]^{4-/3-}$	Couples thermodiffusion in the salt with thermogalvanic redox chemistry; clarifies why very large thermopower may not be purely TG	[8]
Evaporative hydrogel	$[\text{Fe}(\text{CN})_6]^{4-/3-}$ hydrogel	Combines heat harvesting with evaporative cooling; highlights the role of water management in soft TECs	[23]
Strong tough hydrogel	Thermogalvanic hydrogel network	Uses mechanical reinforcement to improve durability while preserving thermoelectric response	[24]

Matrix type	Typical redox/electrolyte choice	Main design function	Main source
Flexible armor-type hydrogel	Thermogalvanic hydrogel device	Emphasizes mechanical protection, flexibility, and high Carnot-relative efficiency for practical form factors	[25]

1.6 Research Gap

Solvent modification provides a direct means of changing the reaction entropy of a redox couple, but the relation between solvent properties, solvation structure and thermopower remains poorly resolved. Thermogalvanic measurements are scarce and are reported for different redox couples, concentrations and experimental conditions. The available dataset is therefore too small for straightforward numerical regression. The problem is compounded by the ion–ion and ion–solvent interactions that determine the local environments of the oxidized and reduced species. These interactions cannot be described reliably by a single solvent parameter. Previous data-driven studies of thermal energy materials have shown the value of machine learning, but they also make clear that a thermogalvanic model must be built around the limited data rather than assuming a large, uniform database [29, 30]. This motivates the use of DFT-derived ion–solvent interaction features and a small-data learning strategy.

Hydrogels add a second level of complexity. Most work on thermogalvanic gels has focused on thermopower, ionic conductivity, mechanical strength, stretchability and stability [24, 31]. These properties are essential for devices, but they do not isolate the role of the polymer in the solvation of the redox couple. The polymer network changes water activity, solvent organization, local polarity and molecular motion. Consequently, the effect of a solvent in a bulk solution cannot be assumed to remain unchanged after gelation. Even the effect of a single solvent or a single polymer on the redox couple is not yet sufficiently understood; varying both solvent and polymer chemistry produces a much larger problem.

The relevant chemical space is at least three-dimensional. The first dimension is the solvent or solvent mixture, which controls polarity, donor–acceptor interactions, viscosity, hydrogen bonding, and redox-ion solubility. The second is the polymer matrix, which introduces functional groups, confinement, heterogeneous water domains, segmental dynamics, and mechanical

constraints. The third is the redox couple itself. Redox couples differ in charge density, coordination geometry, ion pairing, spin state, molecular size, and the accessibility of their solvation shells. A solvent–polymer combination that selectively perturbs ferri/ferrocyanide may therefore have a weak, reversed, or kinetically unfavorable effect on $\text{Fe}^{2+/3+}$, iodide/triiodide, cobalt complexes, or organic redox couples.

These dimensions cannot be optimized independently. A polymer can preferentially retain one solvent component, change the effective solvent composition around the redox species, or create local chemical heterogeneity that is absent from the bulk phase. Conversely, a solvent can plasticize the polymer, disrupt crosslinks, change water retention, or alter mechanical strength. Changing the redox couple then modifies both solvation selectivity and electrochemical compatibility. The resulting combinatorial space is therefore far larger than a list of candidate solvents or polymers, and conventional one-factor-at-a-time screening is unlikely to identify general design rules.

A systematic multi-objective optimization framework has not yet been established for thermogalvanic hydrogels. Maximizing thermopower alone may select strongly interacting environments that increase viscosity, reduce redox-ion mobility, or slow electrode kinetics. Maximizing ionic conductivity alone does not guarantee a large solvation entropy difference or sustained output under load. Likewise, increasing crosslink density can improve strength and dimensional stability while suppressing redox diffusion and solvent reorganization. Practical design must therefore search for Pareto trade-offs among thermopower, ionic and redox-species transport, power density, electrochemical reversibility, thermal stability, water retention, and mechanical performance.

Dynamic thermal response should be included as an independent objective. Steady-state thermopower reports the eventual voltage per unit temperature difference, but it does not reveal how rapidly the device responds to a change in temperature. The voltage rise time, relaxation time, hysteresis, and cycle-to-cycle reproducibility depend on coupled heat transfer, redox-ion diffusion, electrode kinetics, and polymer relaxation. A material with high steady-state thermopower but a slow voltage response may be unsuitable for wearable sensing, intermittent heat sources, or rapidly varying thermal environments. Thermal response time should therefore be reported together with the imposed temperature history and distinguished from purely electrical response time.

CHAPTER 2

AI-INFORMED SOLVATION ENGINEERING OF LIQUID THERMOGALVANIC ELECTROLYTES

2.1 Introduction

Thermogalvanic cells convert a temperature difference into a continuous electrochemical potential through the temperature dependence of a redox reaction. For a simple redox couple in a liquid electrolyte,

$$S_e = \frac{\Delta E}{\Delta T} = \frac{\Delta S_{\text{rc}}}{nF} = \frac{1}{nF} \left(\frac{dG_{\text{solv}}^{\text{O}}}{dT} - \frac{dG_{\text{solv}}^{\text{R}}}{dT} \right), \quad (2.1)$$

where $G_{\text{solv}}^{\text{O}}$ and $G_{\text{solv}}^{\text{R}}$ are the solvation free energies of the oxidized and reduced species. A solvent that perturbs the two redox states unequally can therefore enlarge the solvation entropy difference without changing the electrode architecture [11].

The apparent simplicity of this strategy hides two coupled difficulties. First, thermopower is not controlled by a single solvent property. It emerges from differences in the interaction energies, coordination geometries, packing, orientation, and collective dynamics of solvent molecules around two redox ions. Second, the experimental database is small. The scarcity of S_e measurements for thermogalvanic electrolytes prevents the construction of a large and consistent training set, and the difficulty increases when ion–ion and ion–solvent interactions are included in the prediction. A direct regression of S_e from molecular properties is therefore readily overfitted. Data-driven work on ionic thermoelectrics and thermal-energy materials has reported the same dependence on dataset size and physically relevant features [29, 30].

Here, DFT calculations were combined with machine learning to screen solvent molecules for the $[\text{Fe}(\text{CN})_6]^{4-/3-}$ electrolyte. Ion–solvent interaction features were first parameterized into a descriptor correlated with S_e . The descriptor was then used for high-throughput solvent screening, followed by experimental evaluation and refinement of the model.

2.2 Correlation between Ion–Solvent Interactions and Thermopower

The benchmark system was the aqueous $[\text{Fe}(\text{CN})_6]^{4-/3-}$ redox couple. A useful co-solvent must compete with water for access to the redox ions while creating different local environments around the two charge states. Density functional theory (DFT) calculations were used to quantify ion–solvent interaction energies, bond lengths, and bond angles. These quantities represent interaction strength, the dimensions of the local coordination environment, and the packing or orientation of solvent molecules around the ions, respectively.

For ten solvents with reported thermopower values, the differences in these interaction features between $[\text{Fe}(\text{CN})_6]^{4-}$ and $[\text{Fe}(\text{CN})_6]^{3-}$ were correlated with S_e . No single feature was sufficient, which is consistent with thermopower being a collective solvation property. Partial least-squares dimensionality reduction was consequently used to combine the interaction features into a single descriptor, α . This descriptor correlated strongly with thermopower and varied monotonically with S_e , providing an intermediate target that was more data-efficient and more physically interpretable than a direct solvent-to-thermopower mapping.

The parameterization also revealed an asymmetry between the two redox states. Features associated with the more highly charged $[\text{Fe}(\text{CN})_6]^{4-}$ ion showed stronger correlations with thermopower. Its stronger ion–dipole interactions produce a more ordered solvation shell and make its entropy more sensitive to temperature-induced reorganization. Promising solvents should therefore not simply interact strongly with both ions; they should enlarge the difference between their local responses.

2.3 High-Throughput Solvent Screening

High-throughput calculations generated α values for 64 candidate solvents. Each molecule was represented by ten molecular and physicochemical features: molecular weight, polar surface area, molecular complexity, effective three-dimensional rotor count, density, melting point, viscosity, dielectric constant, dipole moment, and donor number. Dielectric constant, dipole moment, and donor number were included explicitly because of their direct relevance to ion solvation.

The limited dataset made precise numerical regression unstable. The problem was therefore reformulated as interval prediction using a discretized-target gradient boosting classifier (DT-

GBC). A Gaussian mixture model first divided the calculated α values into low, medium, and high classes; a gradient boosting classifier then predicted the class of an unseen solvent. Converting an exact-value problem into a chemically useful ranking problem reduced the effective learning difficulty. The model classified 10 of 13 test samples correctly (76.9% accuracy). Repeated ten-fold cross-validation produced a mean accuracy of 0.64 under changes in the small and imbalanced data splits, quantifying uncertainty that would be hidden by a single train–test partition.

SHAP analysis identified dipole moment as the most influential feature, followed by donor number, polar surface area, and dielectric constant. These features describe complementary aspects of polarity: charge separation, electron donation to an ion, accessibility of polar groups, and dielectric response to charge. Their collective importance established polarity-related features as the most important variables in the model. This result does not imply that thermopower varies monotonically with any one polarity descriptor.

2.4 Experimental Validation and Model Refinement

The initial predictions were tested using six solvents. Triethyl phosphate (TEP), N,N-dimethylformamide (DMF), and ethylenediaminetetraacetic acid followed the predicted trend, whereas glycine, glucose, and erythritol did not. These failures exposed the boundary of the initial representation: it described simple two-body ion–solvent interactions but did not fully capture steric shielding, molecular symmetry, dissociation, or extended hydrogen-bond networks.

The discrepancies were used to add a four-dimensional binary filter based on carbon-chain length, steric hindrance, dissociation energy, and hydrogen-bond donor/acceptor count. The combined procedure correctly classified 15 of 16 evaluated solvents, corresponding to 93.8% accuracy. Thus, the model produced a testable ranking, failed predictions exposed missing physics, and targeted experiments improved the representation.

The refined framework identified TEP, tetrahydrofuran, dimethylacetamide, methylsulfonylmethane, and 3-methyl-2-oxazolidinone as effective candidates, with thermopower enhancements of approximately 42% under the screening conditions. Molecular dynamics simulations further distinguished effective and unsuccessful solvents. Effective molecules such as DMF and TEP produced clearly different solvation structures around the two redox ions, whereas glucose caused only a small difference. The measured thermopower was therefore consistent with asymmetric solvation reorganization rather than solvent polarity alone.

2.5 Solvent Modification of a Thermosensitive Electrolyte

The identified solvents were subsequently evaluated in a guanidinium chloride (GdmCl)-based electrolyte. A 10 wt% GdmCl solution produced $S_e = 2.52 \text{ mV K}^{-1}$. Adding TEP or DMF increased the thermopower to 3.35 and 3.26 mV K^{-1} , respectively. Composition optimization produced a maximum of 4.0 mV K^{-1} at 20 wt% TEP and 20 wt% GdmCl. UV–visible spectra showed that TEP affected $[\text{Fe}(\text{CN})_6]^{4-}$ more strongly than $[\text{Fe}(\text{CN})_6]^{3-}$, consistent with the larger sensitivity of the high-valence ion found in the DFT and molecular-dynamics results.

CHAPTER 3

POLARITY-TUNED HYDROGEL ELECTROLYTES

3.1 Introduction

The liquid-electrolyte study identified polarity-related features as important descriptors of solvent effectiveness, but it did not establish that bulk-liquid relationships remain valid inside a hydrogel. Polymer confinement introduces hydrogen bonding, local dielectric heterogeneity, restricted molecular motion, preferential solvent uptake, and redox-ion transport through a dynamically reorganizing network. These effects can alter both solvation entropy and ionic mobility. The hydrogel study therefore asked whether a solvent polarity rule derived from liquid electrolytes could be converted into a quantitative design principle for a PVA matrix. Derivatives from the dimethyl sulfoxide (DMSO), N,N-dimethylformamide (DMF), and N-methyl-2-pyrrolidone (NMP) families were first compared within PVA hydrogels. Data-driven analysis was then used to identify a cross-family structure–property relationship. The predicted optimum was explored through continuous polarity tuning, and the resulting hydrogel was evaluated from molecular solvation to device-level output.

3.2 Solvent-Family Library in PVA Hydrogels

Within each solvent family, controlled substitution changed molecular properties without replacing the complete chemical scaffold. Across families, the persistence of a trend could be tested against different backbones. Initial structural descriptors generated by cheminformatics tools were sufficiently complex to fit the limited dataset but provided little mechanistic guidance. Isolated HOMO and LUMO energies and electrostatic potential maps likewise did not produce a consistent cross-family relationship with thermopower.

Dipole moment provided a more integrated descriptor. Across the DMSO, DMF, and NMP derivative families, S_e first increased and then decreased with dipole moment, yielding a reproducible volcano-shaped relationship. Weakly polar solvents do not perturb the redox-ion solvation shells sufficiently. At the opposite extreme, strongly polar solvents can over-stabilize and over-order the local environment. Intermediate polarity provides a balance: the solvent

interacts strongly enough to reconstruct the two solvation shells, but not so strongly that both become similarly constrained. The maximum thermopower therefore occurs near an intermediate, rather than maximal, polarity .

3.3 Polarity-Tuned Hydrogel Electrolytes

The discrete set of pure solvents could locate the optimum only coarsely. Continuous tuning was achieved by combining methylsulfonylmethane (MSM) with a lower-polarity co-solvent, denoted PD, in a PVA hydrogel. Varying the PD content shifted the effective polarity through the region surrounding the predicted volcano peak without changing the redox couple or polymer backbone.

Hydrogels containing 4, 6, 8, 10, and 12 vol% PD were evaluated. The maximum thermopower, 3.28 mV K^{-1} , occurred at 8 vol% PD. Lower additions produced values close to the MSM-containing hydrogel, whereas 12 vol% PD reduced performance. This composition dependence confirmed that the optimum was not an artifact of comparing unrelated pure solvents and demonstrated why a monotonic “higher polarity is better” rule would fail.

3.4 Molecular Origin of the Enhanced Thermopower

Cryogenic scanning electron microscopy, solution appearance, and X-ray diffraction indicated that PD created local chemical heterogeneity in the optimized hydrogel. Ab initio molecular dynamics simulations provided a molecular interpretation. In the presence of MSM, PD formed short-range interactions with $[\text{Fe}(\text{CN})_6]^{4-}$, whereas its contribution to the first solvation shell of $[\text{Fe}(\text{CN})_6]^{3-}$ was negligible. The coordination environment of the reduced ion was therefore reorganized more strongly than that of the oxidized ion.

UV–visible and Raman spectroscopy independently supported this asymmetric response. Spectral features associated with $[\text{Fe}(\text{CN})_6]^{4-}$ shifted more strongly upon solvent modification, whereas those of $[\text{Fe}(\text{CN})_6]^{3-}$ changed little. Together, the simulations and experiments suggest that the optimized co-solvent composition preferentially destabilized and reorganized the solvation shell of the more highly charged redox ion. The increase in the difference between the two temperature-dependent solvation free energies connects tuned polarity to enhanced thermopower.

3.5 Thermo-Electrochemical and Dynamic Performance

The optimized composition exposed a trade-off that thermopower alone would miss. Electrochemical impedance spectroscopy showed that the high-frequency real-axis intercept increased from 322 Ω for the MSM hydrogel to 423 Ω for the optimized MSM–PD hydrogel, indicating a modest loss in ionic transport. Nevertheless, the larger thermopower produced a net device benefit. At $\Delta T = 10$ K, the maximum power density increased from 0.24 to 0.44 $\mu\text{W cm}^{-2}$. The optimized thermocell delivered 0.11, 0.25, and 0.44 $\mu\text{W cm}^{-2}$ at temperature differences of 5, 7.5, and 10 K, respectively, while maintaining a normalized power density of approximately 43 $\mu\text{W m}^{-2} \text{K}^{-2}$.

Dynamic measurements were included because steady-state S_e does not describe how quickly a device follows a changing heat source. Under a constant 5 K gradient, the thermocell maintained an output of approximately 18 mV for more than 1.5 h. It also responded reversibly during repeated temperature cycling between 0 and 10 K. With a 5 k Ω load, quasi-continuous charge–discharge cycling caused only a modest voltage decrease from approximately 18 to 15 mV over 30 cycles, and the voltage recovered when the temperature difference was stepped down and returned to 10 K. Thermal-response kinetics and reversibility are therefore independent practical criteria, not consequences that can be inferred from thermopower or ionic conductivity alone.

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